EduFin SQL Business Challenge

Phase 3 — Wednesday Submission

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STEP 3A — The Scale Shift (Comparison Analysis)

```
SELECT
    dc.city_name AS `City`,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END), 2) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2), '%') AS
`Default Rates (%)`
FROM loans_small l
INNER JOIN customers_small c ON l.customer_id = c.customer_id
INNER JOIN dim_city_small dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) DESC
LIMIT 10;

SELECT
    dc.city_name AS `City`,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END), 2) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2), '%') AS
`Default Rates (%)`
FROM loans_national l
INNER JOIN customers_national c ON l.customer_id = c.customer_id
INNER JOIN dim_city_national dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) DESC
LIMIT 10;
```

STEP 3B — CIBIL Score Distribution Among Defaulters

```
WITH cibil_data AS (
    SELECT
```

```
        c.customer_id,
        c.cibil_score,
        c.full_name,
        l.loan_status,
        CASE
            WHEN c.cibil_score > 750 THEN 'Excellent (751+)'
            WHEN c.cibil_score > 700 THEN 'Good (701-750)'
            WHEN c.cibil_score >= 550 THEN 'Fair (550-700)'
            ELSE 'Poor (<550)'
        END AS `CIBIL Brackets`
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
)
SELECT
    `CIBIL Brackets`,
    COUNT(*) AS `Total Loans`,
    SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2), '%') AS
`Default Rate (%)`
FROM cibil_data
GROUP BY `CIBIL Brackets`
ORDER BY SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) DESC;
```

CIBIL Brackets	Total Loans	Defaulted Loans	Default Rate (%)
Fair (550-700)	2390	365	15.27%
Good (701-750)	1214	103	8.48%
Excellent (751+)	1256	83	6.61%
Poor (<550)	140	39	27.86%

STEP 3C — Open Metric Design: "Risk Efficiency"

```
WITH cibil_data AS (
    SELECT
        l.loan_status,
        l.loan_amount,
        CASE
            WHEN c.cibil_score > 750 THEN 'Excellent (751+)'
            WHEN c.cibil_score > 700 THEN 'Good (701-750)'
            WHEN c.cibil_score >= 550 THEN 'Fair (550-700)'
            ELSE 'Poor (<550)'
```

```
        END AS `CIBIL Brackets`
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
)
SELECT
    `CIBIL Brackets`,
    ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount * 1.0 ELSE 0 END) /
    SUM(loan_amount) * LOG(COUNT(*)), 2) AS `Risk Efficiency`
FROM cibil_data
GROUP BY `CIBIL Brackets`
ORDER BY `Risk Efficiency` DESC;
```

STEP 3D — Income and Employment Risk Profile

```
SELECT
    c.employemnet_type AS `Employment Type`,
    CONCAT(ROUND(SUM(l.loan_amount) / 10000000.0, 2), ' Cr') AS `Portfolio Amount`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0 END) / 10000000.0,
2), ' Cr') AS `Defaulted Amount`,
    COUNT(*) AS `Loans`,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Default Count`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2), '%') AS
`Default Rate (%)`
FROM customers c
INNER JOIN loans l
    ON c.customer_id = l.customer_id
GROUP BY c.employemnet_type
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) DESC;
```

Employment Type	Portfolio Amount	Defaulted Amount	Loans	Default Count	Default Rate (%)
Private Employee	94.3 Cr	11.2 Cr	2291	275	12.0%
Self Employed	39.56 Cr	6.91 Cr	967	168	17.37%
Government Employee	39.61 Cr	2.75 Cr	958	62	6.47%
Business Owner	23.5 Cr	2.08 Cr	554	46	8.3%
Student	7.86 Cr	1.3 Cr	230	39	16.96%

STEP 3E — Prioritized Collections Target List (Final Output)

```
WITH defaulted_customer_data AS (
    SELECT
        c.customer_id,
        c.full_name,
        c.cibil_score,
        c.annual_income,
        SUM(l.loan_amount) AS total_loan_amount,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0
                 ELSE 0
            END) AS defaulted_amount,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
    GROUP BY
        c.customer_id,
        c.full_name,
        c.cibil_score,
        c.annual_income
    HAVING SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) > 0
),
risk_score_data AS (
    SELECT
        *,
        ROUND(
            0.5 * (defaulted_amount / 100000.0)
            + 0.3 * default_count * 100000
            + 0.2 * (900 - cibil_score) * 1000,
        2) AS risk_score
    FROM defaulted_customer_data
),
risk_segments AS (
    SELECT
        *,
        AVG(risk_score) OVER () AS avg_risk,
        NTILE(3) OVER (ORDER BY risk_score DESC) AS risk_ntile,
        CASE
            WHEN NTILE(3) OVER (ORDER BY risk_score DESC) = 1 THEN 'High Risk'
            WHEN NTILE(3) OVER (ORDER BY risk_score DESC) = 2 THEN 'Medium Risk'
            ELSE 'Low Risk'
        END AS borrower_segment
    FROM risk_score_data
),
collection_strategy AS (
    SELECT
        *,
        CASE
            WHEN risk_score >= avg_risk THEN 'Above Average Risk'
            ELSE 'Below Average Risk'
        END AS relative_risk,
```

```
        CASE
            WHEN borrower_segment = 'High Risk' AND risk_score >= avg_risk THEN 'Immediate Legal Action'
            WHEN borrower_segment = 'High Risk' THEN 'Escalated Collection'
            WHEN borrower_segment = 'Medium Risk' AND risk_score >= avg_risk THEN 'Legal Notice'
            WHEN borrower_segment = 'Medium Risk' THEN 'Aggressive Collection'
            WHEN borrower_segment = 'Low Risk' AND risk_score >= avg_risk THEN 'Payment Plan Negotiation'
            ELSE 'Standard Recovery'
        END AS collection_approach
    FROM risk_segments
)
SELECT
    customer_id AS `Customer ID`,
    full_name AS `Customer Name`,
    CONCAT(ROUND(total_loan_amount / 100000.0, 2), ' L') AS `Loan Amount`,
    CONCAT(ROUND(defaulted_amount  / 100000.0, 2), ' L') AS `Default Amount`,
    cibil_score AS `CIBIL`,
    CONCAT(ROUND(annual_income     / 100000.0, 2), ' L') AS `Annual Income`,
    default_count AS `Default Count`,
    risk_score AS `Risk Score`,
    borrower_segment AS `Borrower Segment`,
    relative_risk AS `Relative Risk`,
    collection_approach AS `Collection Approach`
FROM collection_strategy
ORDER BY risk_score DESC
LIMIT 50;
```

Customer ID	Customer Name	Loan Amount	Default Amount	CIBIL	Annual Income	Default Count	Risk Score	Borrower Segment	Relative Risk	Collection Approach
1047	Raghav Dubey	15.07 L	15.07 L	486	2.26 L	3	172807.53	High Risk	Above Average Risk	Immediate Legal Action
4724	Ishani Joshi	9.68 L	9.68 L	583	11.33 L	3	153404.84	High Risk	Above Average Risk	Immediate Legal Action
4620	Pushti Doctor	10.28 L	10.28 L	637	5.8 L	3	142605.14	High Risk	Above Average Risk	Immediate Legal Action
1988	Kala Mishra	6.0 L	6.0 L	520	4.2 L	2	136003.0	High Risk	Above Average Risk	Immediate Legal Action
960	Faqid Doshi	9.4 L	9.4 L	534	4.52 L	2	133204.7	High Risk	Above Average Risk	Immediate Legal Action
1087	Oni Lala	5.92 L	5.92 L	535	4.51 L	2	133002.96	High Risk	Above Average Risk	Immediate Legal Action
2463	Hamsini Wadhwa	9.93 L	9.93 L	541	2.46 L	2	131804.96	High Risk	Above Average Risk	Immediate Legal Action
780	Liam Kari	9.14 L	6.98 L	545	3.41 L	2	131003.49	High Risk	Above Average Risk	Immediate Legal Action
4465	Abhimanyu Chana	10.79 L	7.87 L	547	2.88 L	2	130603.94	High Risk	Above Average Risk	Immediate Legal Action
2935	Mohammed Mammen	11.4 L	11.4 L	557	4.22 L	2	128605.7	High Risk	Above Average Risk	Immediate Legal Action
3102	Balveer Swamy	11.06 L	11.06 L	560	1.84 L	2	128005.53	High Risk	Above Average Risk	Immediate Legal Action
3975	Tanay Kuruvilla	13.65 L	9.1 L	571	4.39 L	2	125804.55	High Risk	Above Average Risk	Immediate Legal Action
332	Sai Rama	12.41 L	10.46 L	573	3.46 L	2	125405.23	High Risk	Above Average Risk	Immediate Legal Action
3295	Pooja Sethi	5.3 L	5.3 L	577	0.88 L	2	124602.65	High Risk	Above Average Risk	Immediate Legal Action
992	Jeet Randhawa	9.85 L	9.85 L	580	3.29 L	2	124004.93	High Risk	Above Average Risk	Immediate Legal Action
2079	Qarin Borra	8.42 L	5.1 L	580	12.79 L	2	124002.55	High Risk	Above Average Risk	Immediate Legal Action
3728	Jai Tailor	18.45 L	12.91 L	582	4.17 L	2	123606.46	High Risk	Above Average Risk	Immediate Legal Action
1128	Ronith Anne	10.18 L	10.18 L	582	2.12 L	2	123605.09	High Risk	Above Average Risk	Immediate Legal Action
3198	Fiyaz Talwar	5.12 L	5.12 L	591	3.0 L	2	121802.56	High Risk	Above Average Risk	Immediate Legal Action
804	Parth Reddy	9.81 L	5.63 L	592	2.27 L	2	121602.82	High Risk	Above Average Risk	Immediate Legal Action
3298	Yashasvi Sundaram	17.46 L	12.97 L	598	7.4 L	2	120406.48	High Risk	Above Average Risk	Immediate Legal Action
1619	Riya Pathak	10.8 L	4.48 L	602	8.42 L	2	119602.24	High Risk	Above Average Risk	Immediate Legal Action
29	Ganga Kala	7.68 L	7.68 L	605	11.18 L	2	119003.84	High Risk	Above Average Risk	Immediate Legal Action
3041	Bishakha Tella	8.13 L	8.13 L	627	6.36 L	2	114604.06	High Risk	Above Average Risk	Immediate Legal Action
4431	Amol Nayak	8.74 L	8.74 L	630	4.51 L	2	114004.37	High Risk	Above Average Risk	Immediate Legal Action
4900	Imaran Sachar	13.89 L	2.89 L	485	4.12 L	1	113001.44	High Risk	Above Average Risk	Immediate Legal Action
872	Madhav Padmanabhan	5.56 L	5.56 L	641	1.02 L	2	111802.78	High Risk	Above Average Risk	Immediate Legal Action

Custom Column

```
WITH defaulted_customer_data AS (
    SELECT
        c.customer_id,
        c.full_name,
        c.cibil_score,
        c.annual_income,
        SUM(l.loan_amount) AS total_loan_amount,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0 END) AS defaulted_amount,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
    GROUP BY c.customer_id, c.full_name, c.cibil_score, c.annual_income
    HAVING SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) > 0
),
risk_score_data AS (
    SELECT
        *,
        ROUND(
            0.5 * (defaulted_amount / 100000.0) +
            0.3 * default_count * 100000 +
            0.2 * (900 - cibil_score) * 1000    ,
        2) AS risk_score
    FROM defaulted_customer_data
),
risk_segments AS (
    SELECT
        *,
        AVG(risk_score) OVER () AS avg_risk,
        NTILE(3) OVER (ORDER BY risk_score DESC) AS risk_ntile,
        CASE
            WHEN NTILE(3) OVER (ORDER BY risk_score DESC) = 1 THEN 'High Risk'
            WHEN NTILE(3) OVER (ORDER BY risk_score DESC) = 2 THEN 'Medium Risk'
            ELSE 'Low Risk'
        END AS borrower_segment
    FROM risk_score_data
),
collection_strategy AS (
    SELECT
        *,
        CASE
            WHEN risk_score >= avg_risk THEN 'Above Average Risk'
            ELSE 'Below Average Risk'
        END AS relative_risk,
        CASE
            WHEN borrower_segment = 'High Risk' AND risk_score >= avg_risk
                THEN 'Immediate Legal Action'
            WHEN borrower_segment = 'High Risk'
```

```
                THEN 'Escalated Collection'
            WHEN borrower_segment = 'Medium Risk' AND risk_score >= avg_risk
                THEN 'Legal Notice'
            WHEN borrower_segment = 'Medium Risk'
                THEN 'Aggressive Collection'
            WHEN borrower_segment = 'Low Risk' AND risk_score >= avg_risk
                THEN 'Payment Plan Negotiation'
            ELSE 'Standard Recovery'
        END AS collection_approach
    FROM risk_segments
)
SELECT
    customer_id AS `Customer ID`,
    full_name AS `Customer Name`,
    CONCAT(ROUND(total_loan_amount / 100000.0, 2), ' L') AS `Loan Amount`,
    CONCAT(ROUND(defaulted_amount / 100000.0, 2), ' L') AS `Default Amount`,
    cibil_score AS `CIBIL`,
    CONCAT(ROUND(annual_income / 100000.0, 2), ' L') AS `Annual Income`,
    default_count AS `Default Count`,
    risk_score AS `Risk Score`,
    borrower_segment AS `Borrower Segment`,
    relative_risk AS `Relative Risk`,
    collection_approach AS `Collection Approach`
FROM collection_strategy
ORDER BY risk_score DESC;

CASE
  WHEN borrower_segment = 'High Risk' AND risk_score >= avg_risk
      THEN 'Immediate Legal Action'
  WHEN borrower_segment = 'High Risk'
      THEN 'Escalated Collection'
  WHEN borrower_segment = 'Medium Risk' AND risk_score >= avg_risk
      THEN 'Legal Notice'
  WHEN borrower_segment = 'Medium Risk'
      THEN 'Aggressive Collection'
  WHEN borrower_segment = 'Low Risk' AND risk_score >= avg_risk
      THEN 'Payment Plan Negotiation'
  ELSE 'Standard Recovery'
END AS collection_approach
```

> CTEs are better because they break complex logic into readable, named stages rather than burying it in nested CASE statements. This makes debugging easier (you can inspect each CTE layer) and lets the database reuse intermediate results instead of recalculating them.

WHY Gate

1. If you added employment_type as a third factor in borrower_segment — would any customers move to a different segment? Pick one specific customer and show the re-classification logic.

> In my current logic to calculating borrower segment, i have used the risk score (default_amount, default_count & annual income) and not CIBIL brackets. When employment_type is added, it creates a composite_risk_score through a weighted formula, which changes the customer ranking order, customers with better employment stability get lower composite scores and may shift from High Risk to Medium Risk if their new rank crosses the NTILE boundary.

2. Your top 50 list: the collections team can make 200 calls per week. How do you prioritise the order within your top 50? Is priority_score alone sufficient?

> I have ordered my top 50 list using the risk score which was calculated using weighted formula which included default_count, default_amount and annual_income of each custoemrs. This way of ordering is sufficient because for 200 calls/week on a top 50 list, the collections team needs customers ranked by who will recover the most money with highest likelihood, which is exactly what our risk_score is capturing: larger defaults (50% weight) matter most, repeat defaulters (30% weight) are flagged second, and credit quality (20% weight) adds predictability. The ranking already prioritizes financial exposure + behavioral risk, so the team can work top-to-bottom without needing additional tie-breakers.

3. Write the "Riskiest Borrower Profile" paragraph. Describe the type of customer who appears most frequently in your top segment.

> The riskiest borrower profile is typically a self-employed or student customer with a low to mid CIBIL score and multiple past defaults. These borrowers often have higher defaulted loan amounts and repeated default behavior, indicating weak repayment stability. They appear most frequently in the top risk segment due to both financial exposure and unstable income sources.

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